

Created files:

Convolution.sv

Convolution_tb.sv

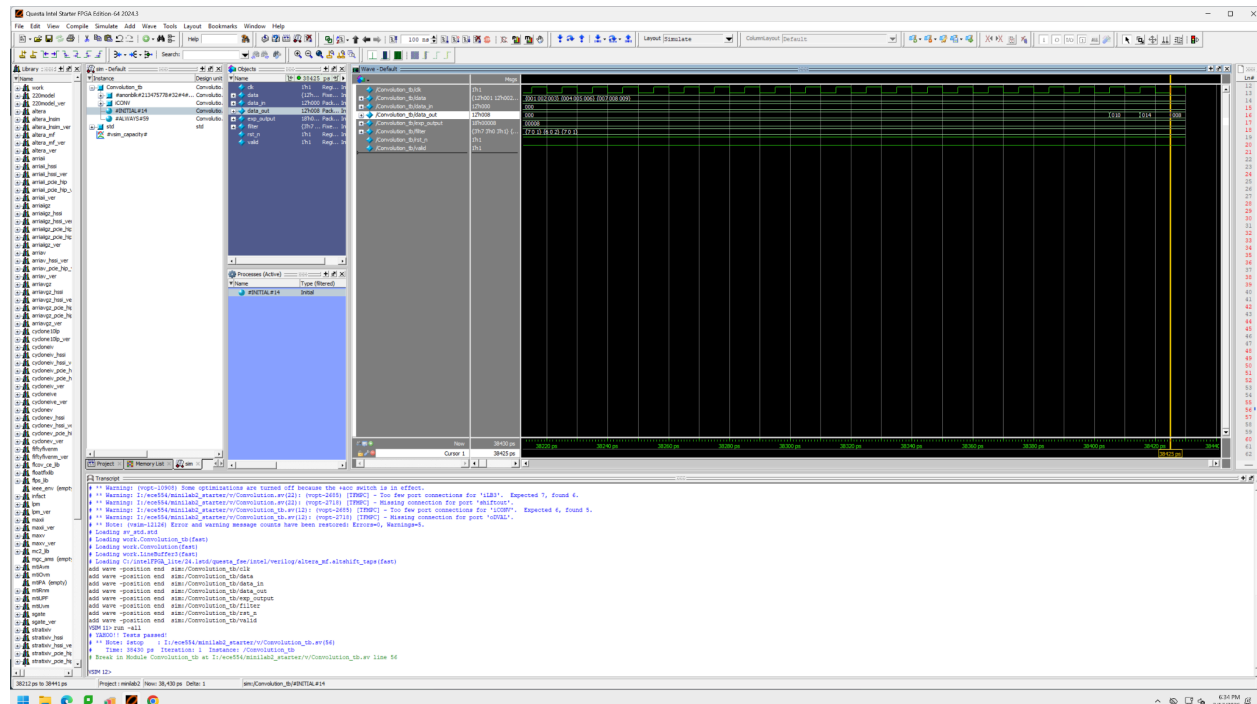
to_grayscale.sv

to_grayscale_tb.sv

DE1_SoC_CAMERA.v (MODIFIED)

Minilab2 Report

In our implementation we have a grayscale module that takes 4 pixels that have been shifted out of 2 taps and into 2 registers then combines each 12 bit value and then divides by 4. We selfchecked this module with a test bench that ran for 2480 cycles to get enough data values out of the taps. Then there is a convolution module that takes three rows of pixels and multiplies each square by a sobel filter which is a parameter to create both a horizontal and vertical edge detection. The horizontal kernel is our default parameter(aka vertical edge detection). Our test bench provided 3 values every 1280 cycles for the taps to provide actual data out of each tap and then checks that after 3840 cycles the value predicted is correct.



In our top level file we made gray scale our output whenever no switches were on, colored image whenever switch 1 was on, and when switch 2 was on we were doing convolution. But we had actually just instantiated two convolution modules one with the default parameter and one with the input filter as the vertical kernel. So then whenever switch two was on if switch 3 was also on we were getting horizontal edge detection otherwise when switch 3 was off we had vertical edge detection.

We ran into an issue where our camera image was being duplicated as well as what was supposed to be the left and right side of the image was overlaid. We determined that it was

because our taps distance was 640 causing it to display the right side of the image to display at the same point as the left side of the image. even though the resolution ends up being 640 the image rows are still coming in at 1280.

Utilization Report

3% of ALMs were used just from our basic logic. 4% of total block memory bits(BRAMs) was used most likely from the line buffers. There were no DSPs being reported meaning most of our math is being done in ALMs. Ultimately it was a relatively small design mainly consisted of storing rows of pixels.